

WEEK:- 14

Topic: Groq Cloud AI, Prompt Engineering, and AI-Based Structured Data Extraction

Objectives:

To understand Groq Cloud AI services, learn prompt engineering techniques, and develop AI-powered solutions for extracting structured information from unstructured data sources.

Learning Outcomes:

- Learned the fundamentals of Groq Cloud AI and its applications.
- Understood prompt engineering techniques for AI model interactions.
- Gained knowledge of structured data extraction using AI technologies.
- Developed skills in designing effective prompts for accurate results.
- Improved understanding of AI-driven automation and data processing workflows.

WORK SUMMARY:

During Week 14, I worked on Groq Cloud AI, prompt engineering, and AI-based structured data extraction. The training focused on understanding how artificial intelligence can be utilized to process unstructured information and transform it into meaningful, structured data. I attended technical sessions and participated in practical exercises to explore AI-powered solutions and their applications in business process automation. Various use cases were studied to understand how AI models interpret instructions and generate accurate outputs based on provided prompts.

Throughout the week, I gained practical experience in working with Groq Cloud AI services and designing effective prompts to improve response quality and accuracy. I learned how prompt structure, context, and instruction design influence AI-generated results. Special emphasis was placed on extracting structured information from documents, invoices, and other business records using AI models. I also explored techniques for refining prompts, validating outputs, and ensuring data consistency across different extraction scenarios.

The activities carried out during this week enhanced my understanding of AI technologies and their practical implementation in real-world applications. Regular experimentation, testing, and mentor feedback helped me develop analytical thinking and optimization skills. The knowledge and practical experience gained during this week strengthened my understanding of intelligent automation and provided valuable insights into modern AI-driven data processing solutions.

The knowledge gained during this week contributed significantly to the development of the final project and helped me understand professional software development environments. Regular reviews, troubleshooting exercises, and practical demonstrations provided valuable learning opportunities and improved my confidence in working with modern technologies.

Activities Performed:

- Studied Groq Cloud AI services and implemented prompt engineering techniques.
- Performed AI-based structured data extraction and output validation.